International Journal of Computing and Digital Systems
2024, VOL. 17, NO. 1, 1–10
http://dx.doi.org/10.12785/ijcds/XXXXXX

Improving Tree Structure with SR-CCP: A Regularized Pruning Strategy for Shallower and Simpler Decision Models

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Received 12 March 2024, Revised 15 June 2024, Accepted 10 July 2024

Abstract: Balancing predictive performance and model simplicity remains a central challenge in classification tasks using decision trees. Despite their widespread adoption in machine learning for their interpretability and efficiency, decision trees are prone to excessive growth when trained on noisy or high-dimensional data sets, often resulting in overfitting and structural complexity. Traditional Cost-Complexity Pruning (CCP) addresses this issue by penalizing the number of terminal nodes, but it fails to consider additional structural aspects such as tree depth and feature usage. This work suggests a novel and potential technique to get beyond these restrictions, called Structural Regularized Cost-Complexity Pruning (SR-CCP), as an extension of CCP that incorporates explicit penalties on depth and the proportion of features used during pruning. The method is evaluated in seven benchmark data sets using stratified cross-validation, comparing its performance against five established pruning strategies. Evaluation metrics include classification accuracy, F1 score, tree depth, number of nodes, leaves, usage of features, and inference time. Experimental results demonstrated that SR-CCP consistently maintains competitive predictive performance across all pruning methods, preserving comparable accuracy and F1 score values. In addition, the proposed method achieved substantially greater structural parsimony, producing significantly more compact trees characterized by reduced depth, fewer internal nodes, and a lower number of terminal leaves compared to all other variants of the pruning model.

Keywords: Decision Tree, Classification, Pruning, Cost-Complexity Pruning, Structural Regularization.

1. Introduction

Decision trees remain a common choice in machine learning due to their ease of interpretation, model transparency, and relatively low computational cost for classification and regression tasks [1]. However, when a model is trained on noisy or high-dimensional data sets, it tends to grow excessively in size, resulting in complex and overly deep structures that are difficult to interpret and may not generalize well to unseen data; overfitting is a prime example of poor generalization, where the model becomes excessively tailored to the training set [2]. These issues become particularly problematic when the model captures spurious patterns that do not contribute to predictive accuracy, leading to unnecessarily complex trees with reduced performance on new data [3].

In recent years, extensive evaluations of pre- and post-pruning techniques have been conducted in various medical data sets [4] demonstrating that the effectiveness of pruning depends largely on the characteristics of the data set. Furthermore, despite the development of numerous variants such as Iterative Dichotomiser 3 (ID3), which generates trees using information gain as a splitting criterion, C4.5, which incorporates gain ratio and handles continuous attributes and missing values at the same time; and Classification and Regression Trees (CART), which supports classification and regression tasks using Gini impurity or variance reduction as splitting rules, decision trees still face challenges related to structural complexity, interpretability, and inference time. This reinforces the need for pruning methods that explicitly consider structural metrics in their formulation [5].

However, other methods include Reduced Error Pruning (REP), which prunes subtrees without reducing the validation error. Minimal Error Pruning (MEP), which estimates classification error before and after pruning a node [6]. And Depth-Based Pruning (DBP), a pre-pruning algorithm that limits tree growth to a maximum depth. Although these methods can improve generalization or reduce size, they often lack mechanisms to control structural complexity metrics, such as depth or feature number, in a principled way.

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Although the more robust and fundamental pruning technique is Cost-Complexity Pruning (CCP) [7], which uses a complexity penalty to balance simplicity and precision. The pruning process generates a sequence of subtrees as the original tree is repeatedly pruned, with each contributing the least to predictive ability relative to complexity cost. Although effective, a major limitation of CCP remains the sensitivity of the regularization parameter and its inability to account for other structural concerns, such as depth or feature usage. In addition, the depth of the tree and the incorporation of numerous features can substantially affect the ability of the model to generalize, even when the number of leaf nodes is limited [8]. Furthermore, previous work has viewed the task of minimizing tree size as a combinatorial optimization problem [9], which motivates the integration of explicit structural regularization in pruning strategies.

Likewise, variants of CCP [10] have been studied and extended for ensembles such as Random Forests, leveraging out-of-bag samples to optimize tree size and generalization, where excessive depth or size can impair explainability and marginally reduce accuracy, underscoring the ongoing need to better control structural complexity during pruning.

For that reason, this paper proposes a novel pruning method, called Structural Regularized Cost-Complexity Pruning (SR-CCP), which incorporates structural penalties on tree depth and feature usage into the classical cost-complexity pruning framework. The proposed method aims to generate more compact and simpler decision trees while maintaining similar predictive performance. This approach could be of significant value in domains where model interpretability is critical, such as healthcare, finance, and security, where transparent and efficient decision-making models are essential.

The remainder of the structure of the paper is organized in the following way: Section 2 offers a review of related work on pruning of decision trees and structural regularization, Section 3 describes the proposed SR-CCP method including the formulation and optimization process, Section 4 shows the experimental results along with their respective analysis, and the conclusions are shown in Section 5.

2. Related Work

Over the years, numerous pruning techniques have been developed to address the structural complexity and generalization issues inherent in decision trees. These methods vary in their underlying assumptions, pruning criteria, and empirical performance across domains. This section reviews the most influential approaches, highlighting their strengths and limitations, particularly in terms of model simplicity, predictive reliability, and computational cost. One of the earliest methods was introduced by Quinlan [11], it was called Reduced Error Pruning (REP), which prunes away any subtrees if doing so improves performance on a validation set. While it can significantly reduce tree structure, REP can also prune relevant information in the event of dealing with a noisy data set. Bohanec and Bratko [12] proposed an alternative pruning technique called Error-Based Pruning (EBP), which guides pruning by fair estimation of both node-specific and total classification errors. Although EBP was able to reduce the tree size by 30% while maintaining accuracy relative to the entire decision tree, it struggled to generate local decisions when the data sets were unbalanced with respect to classes, providing limited generalization.

Consequently, Yang et al. [13] introduced an enhanced EBP algorithm for large data sets, generating small trees with acceptable accuracy levels. However, performance still varied for trees with highly imbalanced structures. Additionally, Basgalupp et al. [14] proposed a multi-objective genetic algorithm to balance predictive accuracy and model interpretability, albeit at the cost of increased computational complexity.

The MEP approach that originated with the work of Niblett and Bratko was later improved by the contributions of Cestnik through the incorporation of a Bayesian probability estimation method based on m-estimates [15]. Unlike classical procedures such as Laplace estimation, MEP introduces a tunable parameter to adjust the weight of prior probabilities relative to empirical data. This flexibility enables domain-specific adaptation, especially in noisy domains, while also addressing class imbalance more efficiently. Furthermore, MEP supports pruning at varying levels of complexity, allowing for finer control over tree simplification. In a comparative study, Mingers [6] evaluated MEP against alternative pruning methods and found that its parameter-based optimization improved the model's robustness across various scenarios. An additional pruning strategy is Depth-Based Pruning (DBP), where tree growth is stopped at a predetermined maximum depth. This method enforces a hard constraint on the tree's structural size, and it is especially effective in reducing computational costs, albeit at the risk of underfitting if the depth limit is too conservative. Messmer and Bunke [16] introduced a variant of this principle in the context of graph isomorphism detection, where they constructed large decision trees for subgraph matching and applied pruning mechanisms to minimize depth and redundancy. Their strategy demonstrates how tree depth limitations can serve as an effective structural control mechanism, particularly in scenarios involving large model spaces and real-time requirements.

From a domain perspective, Amro et al. [2] further highlighted the risk of overfitting in clinical decision systems where interpretability is as important as accuracy. Likewise, Shamrat et al. [4] showed extensive experimental results of pre-pruning and post-pruning against several medical data sets, but showed how variability in the data set itself can heavily influence the potential effectiveness of the post-pruning procedure.

Also, comparative studies [17] confirm that there is no universally best pruning method. However, the studies of Kotsiantis [18] and Liu et al. [19] demonstrated that feature discretization and data set dimensionality are important factors in practical or theoretical work in different kinds of pruning techniques.

A major recent contribution to pruning theory comes from Harviainen et al. [20], who reconsidered optimal decision tree pruning in a computational complexity context. Their contribution is a complete classification of all pruning operations out of two basic modifications (subtree exchange, subtree elevation). They showed that while optimal subtree exchange is tractable, subtree elevation is NP-complete, even in restricted forms. Furthermore, they presented dynamic programming algorithms and complexity reasoning that show that theoretical limits of fully optimal pruning exist.

In summary, while existing pruning methods have demonstrated distinct advances in reducing tree complexity and improving generalization, they often lack principled mechanisms to directly control structural aspects such as depth and feature usage. These limitations motivate the development of our proposed SR-CCP method, which introduces explicit structural penalties into the pruning process to enhance both interpretability and predictive performance.

3. Proposed Method

Figure 1 illustrates the proposed framework in six stages, detailing the data set, model training, hyperparameter optimization, and evaluation procedures.

A. Data sets

In this study, seven public data sets were used. Table I presents the main characteristics of them. These data sets span multiple domains, including security, health, chemistry, and software engineering, and were selected to ensure variability in classification difficulty and structural tree complexity.

Each data set was obtained from either the *UCI Machine Learning Repository* [21] or *OpenML* [22], and each was preprocessed. In cases where the target variable was categorical, an encoding was assigned to map the class labels to integer values in a deterministic and consistent manner. No missing values were found in any of the data sets, so no additional transformation was required. Consequently, all data sets were treated as clean, numerical, and binary classification tasks, allowing for a fair and consistent comparison between pruning strategies.

The Table I shows the chosen data sets with the number of samples, features, classes, and indicates whether they are balanced or not. The most complex database is SpamBase.

B. Decision Tree Variants

This study compares six variants of decision trees, including the proposed method. These methods include standard pruning techniques as well as pre-pruning and structurally penalized strategies.

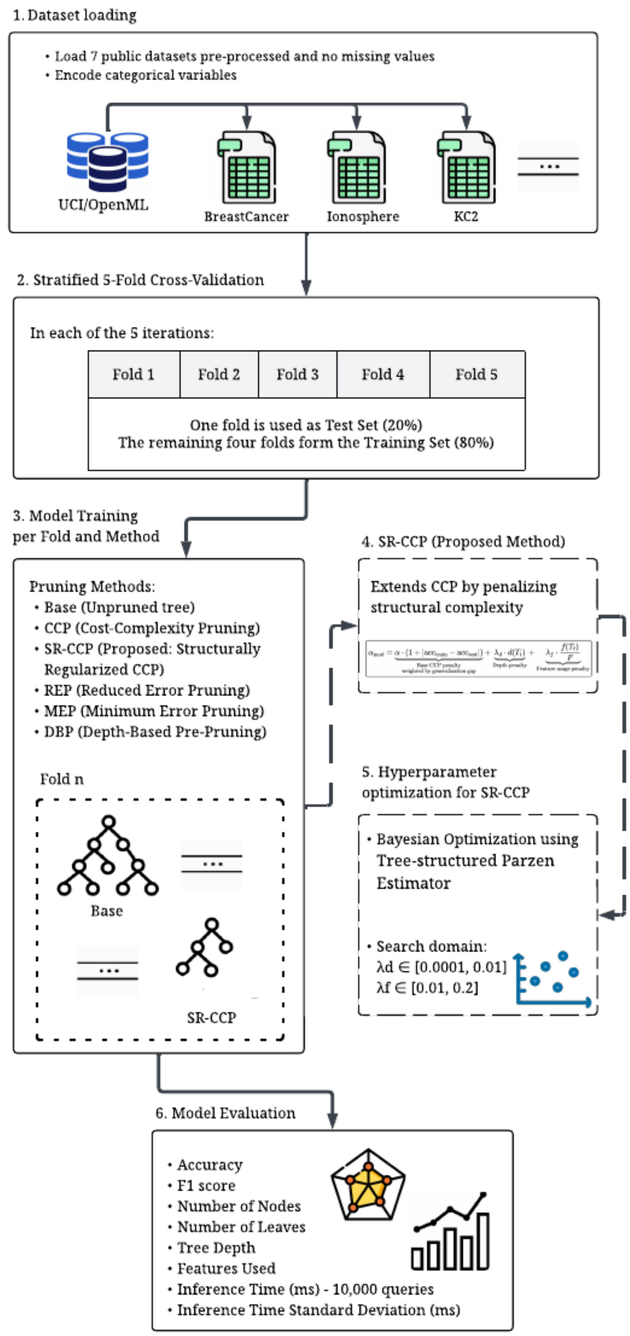

Figure 1. Flowchart of the proposed method as an extension of CCP that incorporates explicit penalties during the pruning process.

TABLE I. Summary of the data sets used in the experimental evaluation

Data set	Domain	Samples	Features	Classes	Balanced
BanknoteAuth	Security	1372	4	2	Yes
BreastCancer	Health	569	30	2	No
HeartDisease	Health	270	13	2	Yes
Ionosphere	Signal Processing	351	34	2	No
KC2	Software Engineering	522	21	2	No
QSARBiodeg	Chemistry	1055	41	2	No
SpamBase	Security	4601	57	2	Yes

1) Baseline Tree

The baseline model is a fully grown decision tree trained with the Gini impurity index [23]. See equation 1.

$$G(t) = 1 - \sum_{i=1}^{C} p_i^2(t) \tag{1}$$

where C represents the number of classes, and $p_i(t)$ denotes the proportion of instances of class i at node t.

2) Cost-Complexity Pruning

CCP prunes a subtree by minimizing a regularized risk, using the equation 2.

$$R_\alpha(T) = R(T) + \alpha \cdot |\text{leaves}(T)| \tag{2}$$

Where $R(T)$ is the empirical error and α is a regularization coefficient that balances fit and complexity [7].

3) Proposed Structurally Regularized CCP

This is our proposed pruning technique, called SR-CCP, which incorporates structural penalties in classical CCP, based on the depth and proportion of features used, promoting the formation of simpler trees. Concretely, SR-CCP modifies the CCP cost function as expressed in equation 6, where total cost combines prediction error penalty, depth penalty and feature utilization penalty. Its formulation is detailed in equation 6.

4) Reduced Error Pruning

REP is a post-pruning strategy that replaces subtrees with majority-class leaves if doing so does not reduce validation accuracy. See equation 3.

$$\text{Accuracy}(T', D_{\text{val}}) \geq \text{Accuracy}(T, D_{\text{val}}) \tag{3}$$

5) Minimum Error Pruning

MEP compares the estimated classification error of a node considered as a leaf versus its current subtree [24]. A node is pruned if it satisfies equation 4.

$$\hat{e}(\ell_t) = \frac{n_t - n_{\max}}{n_t} \leq \hat{e}(T_t) \tag{4}$$

Where n_t is the number of samples at node t, and $n_{\max}$ is the count of the majority class in that node.

6) Depth-Based Pruning (DBP)

DBP is a pre-pruning technique that halts tree growth at a fixed depth $d_{\max}$ [16]. See equation 5.

$$\text{Depth}(T) \leq d_{\max} \tag{5}$$

C. Proposed pruning technique

1) Structurally Regularized Cost-Complexity Pruning

The SR-CCP is a novel cost-complexity pruning strategy proposed in this study, which extends the classical CCP by incorporating structural penalties to better control tree complexity. Instead of relying solely on the empirical trade-off between accuracy and tree size, SR-CCP adjusts the base complexity parameter α by considering both the generalization gap, and structural attributes of the candidate subtrees. The modified pruning coefficient is defined in the equation 6.

$$\alpha_{\text{mod}} = \alpha \cdot (1 + |\text{acc}_{\text{train}} - \text{acc}_{\text{test}}|) + \lambda_d \cdot d(T_t) + \lambda_f \cdot \frac{f(T_t)}{F} \tag{6}$$

Where $\alpha \in \mathbb{R}_{\geq 0}$ denotes the original CCP complexity parameter. The terms $\text{acc}_{\text{train}}$ and acc_{test} are the classification accuracy in the training and validation data sets, which captures the generalization performance of the tree. The structural complexity is explicitly controlled through two additional components: the depth $d(T_t) \in \mathbb{N}$ of the candidate subtree T_t, and the number of distinct features $f(T_t) \in \mathbb{N}$ used in T_t, normalized by the total number of available features $F \in \mathbb{N}$. The hyperparameters $\lambda_d, \lambda_f \in \mathbb{R}_{\geq 0}$ define the penalty level associated with the depth and feature used, respectively. By integrating these terms, SR-CCP aims to generate more interpretable models without compromising predictive performance too much.

The proposed SR-CCP procedure is detailed in Algorithm 1.

Algorithm 1 Proposed Structurally Regularized CCP

Input: Training set D_{train}, validation set D_{val}, regularization weights λ_d, λ_f
Output: Pruned tree T^*
Compute CCP pruning path $\{\alpha_0, \alpha_1, \ldots, \alpha_k\}$ from D_{train}
Initialize best_acc $\leftarrow -\infty$, $T^* \leftarrow \emptyset$
for each $\alpha \in \{\alpha_0, \ldots, \alpha_k\}$ **do**
  Train $T \leftarrow$ DecisionTree($ccp\_alpha = \alpha$) on D_{train}
  Compute $\text{acc}_{\text{train}} \leftarrow \text{Accuracy}(T, D_{\text{train}})$
  Compute $\text{acc}_{\text{val}} \leftarrow \text{Accuracy}(T, D_{\text{val}})$
  Compute depth $\leftarrow$ TreeDepth(T)
  Compute $\text{features}_{\text{used}} \leftarrow \text{FeaturesUsed}(T)$
  Compute $\alpha_{\text{mod}} \leftarrow \alpha \cdot (1 + |\text{acc}_{\text{train}} - \text{acc}_{\text{val}}|)$
    $+\lambda_d \cdot \text{depth} + \lambda_f \cdot \left(\frac{\text{features}_{\text{used}}}{\text{features}_{\text{total}}}\right)$
  Retrain $T_{\text{mod}} \leftarrow$ DecisionTree($ccp\_alpha = \alpha_{\text{mod}}$) on D_{train}
  Compute $\text{acc}_{\text{mod}} \leftarrow \text{Accuracy}(T_{\text{mod}}, D_{\text{val}})$
  if $\text{acc}_{\text{mod}} >$ best_acc **then**
    Update $T^* \leftarrow T_{\text{mod}}$
    Update best_acc $\leftarrow \text{acc}_{\text{mod}}$
  end if
end for
Return T^*

D. *Hyperparameter Optimization*

The regularization parameters λ_d and λ_f are tuned using Bayesian Optimization with the Tree-structured Parzen Estimator (TPE) algorithm [25], whose search space is defined in equation 7 .

$$\lambda_d \in [0.00001, 0.01], \quad \lambda_f \in [0.01, 0.2] \tag{7}$$

The goal is to maximize validation accuracy. The procedure is described in Algorithm 2.

Algorithm 2 Bayesian Hyperparameter Optimization for SR-CCP

Input: Training set D_{train}, validation set D_{val}, search space $\Lambda = \{(\lambda_d, \lambda_f) \mid \lambda_d \in [\lambda_d^{\min}, \lambda_d^{\max}], \lambda_f \in [\lambda_f^{\min}, \lambda_f^{\max}]\}$, maximum trials N
Output: Optimal hyperparameters $(\lambda_d^*, \lambda_f^*)$
Initialize Bayesian optimizer surrogate model $\mathcal{P}$
Initialize best_acc $\leftarrow -\infty$
for $i = 1$ to N **do**
  Sample $(\lambda_d, \lambda_f) \sim \text{AcquisitionFunction}(\mathcal{P})$
  $T \leftarrow \text{SR-CCP}(D_{\text{train}}, D_{\text{val}}, \lambda_d, \lambda_f)$
  Compute acc $\leftarrow \text{Accuracy}(T, D_{\text{val}})$
  if acc $>$ best_acc **then**
    Update best_acc $\leftarrow$ acc
    Update $(\lambda_d^*, \lambda_f^*) \leftarrow (\lambda_d, \lambda_f)$
  end if
  Update surrogate model $\mathcal{P}$ with $(\lambda_d, \lambda_f, \text{acc})$
end for
Return $(\lambda_d^*, \lambda_f^*)$

E. *Evaluation*

To obtain reliable and unbiased performance estimates, each pruning method was tested with a 5-fold stratified cross-validation procedure. Within each fold, models were trained and evaluated, preserving the original class distribution. For our proposed SR-CCP, hyperparameter optimization was performed during the training phase using internal validation.

The following metrics were recorded for each model: classification accuracy, F1 score, tree depth, number of internal nodes, number of terminal leaves, number of features used in decision splits, and inference time. This last was measured over 10,000 repetitions (excluding the first as warm-up), reporting both the median and standard deviation to mitigate the effects of transient system variability.

The structural metrics evaluated the complexity and interpretability of the model, while the performance metrics measured predictive effectiveness. The experiments were conducted using a system with an Intel® CoreTM i7-13620H CPU, 16 GB RAM, and Ubuntu 24.04, without GPU acceleration or parallel processing.

4. Experiments and Results

The effectiveness of the proposed method was measured in terms of accuracy and F1 score, five decision tree pruning strategies were implemented. In addition, five comparative characteristics were considered: number of leaves, total nodes, depth, number of features, and inference time in milliseconds with its standard deviation (10,000 runs). This complete multidimensional evaluation of each pruning method in predictive performance and structural simplification is shown in Table II.

A. *Predictive Performance*

We used seven data sets, whose characteristics were described in Table I, to assess how well our proposed SR-CCP method performs.

Table II displays the performance metrics (accuracy and F1 score), with values shown as the mean from stratified cross-validation. The higher values are marked in bold in the table. As expected, the accuracy and F1 score of our proposed method show the best performance, as detailed in Table III, which includes the mean and median of each metric across the six data sets, except for Median for F1 score, whose value is the second-highest value.

1) *Accuracy Analysis*

Based on the results shown in Table II, the accuracy performance was compared between the seven data sets considered in this study.

The proposed SR-CCP method achieved the highest overall predictive accuracy, with a mean of 89.47% and a median of 91.65%, followed closely by CCP (89.22% mean, 91.17% median) and MEP (88.81% mean, 91.48% median). The unpruned Base model yielded lower average

TABLE II. Summary of Accuracy, F1 score, Leaves, Nodes, Depth, Features, and Inference Time (median ± std) for pruning methods

Data set	Method	Accuracy	F1 score	Leaves	Nodes	Depth	Features	Inference Time (ms)
BanknoteAuth	Base	0.98323	0.98303	27.6	54.2	8.8	4	0.06566 ± 0.00285
	CCP	0.98688	0.98671	23.4	45.8	8.2	4	0.06568 ± 0.00263
	Proposed SR-CCP	0.98742	0.98727	**8.4**	**15.8**	**5.2**	**3**	**0.06437 ± 0.00243**
	REP	0.98469	0.98451	14.8	28.6	7.0	4	0.06513 ± 0.00343
	MEP	**0.98761**	**0.98746**	15.4	29.8	7.4	4	0.06504 ± 0.00255
	DBP	0.97085	0.97042	18.8	36.6	6.0	4	0.06519 ± 0.00236
BreastCancer	Base	0.91040	0.90279	18.8	36.6	8.2	12.6	0.06282 ± 0.00342
	CCP	0.93149	0.92551	11.8	22.6	6.2	7.8	0.06248 ± 0.00269
	Proposed SR-CCP	**0.94377**	**0.93960**	**3.2**	**5.4**	**3.0**	**2.2**	**0.06223 ± 0.00260**
	REP	0.93145	0.92610	5.2	9.4	4.8	5	0.06228 ± 0.00265
	MEP	0.93322	0.92797	5.4	9.8	5.0	5.2	0.06260 ± 0.00256
	DBP	0.92799	0.92150	14.2	27.4	6.0	9.8	0.06252 ± 0.00237
HeartDisease	Base	0.72593	0.72199	37.0	73.0	9.2	11.4	0.05944 ± 0.00317
	CCP	0.78519	0.78146	25.6	50.2	7.8	9.8	0.05942 ± 0.00322
	Proposed SR-CCP	**0.79630**	**0.79203**	**6.0**	**11.0**	4.4	**4**	**0.05909 ± 0.00272**
	REP	0.75556	0.73687	6.4	11.8	**4.2**	6.4	0.05924 ± 0.00245
	MEP	0.77407	0.75384	7.4	13.8	4.8	6.4	0.05936 ± 0.00210
	DBP	0.73704	0.73180	24.2	47.4	6.0	10.6	0.05939 ± 0.00247
Ionosphere	Base	0.89742	0.88878	20.2	39.4	10.2	14.6	0.06112 ± 0.00323
	CCP	0.91167	0.90360	12.6	24.2	7.6	9.4	0.06098 ± 0.00218
	Proposed SR-CCP	0.91653	0.90618	**3.4**	**5.8**	**3.4**	**2.4**	**0.06086 ± 0.00220**
	REP	0.91167	0.90412	6.0	11.0	5.4	5.6	0.06089 ± 0.00256
	MEP	**0.92024**	**0.91339**	7.2	13.4	6.4	6.8	0.06094 ± 0.00199
	DBP	0.89738	0.88544	12.8	24.6	6.0	9.8	0.06098 ± 0.00219
KC2	Base	0.81806	0.71764	57.4	113.8	14.0	19.2	0.06344 ± 0.00296
	CCP	**0.86015**	**0.77822**	15.8	30.6	6.6	6.8	0.06248 ± 0.00265
	Proposed SR-CCP	0.85060	0.77328	**2.0**	**3.0**	**2.0**	**1**	**0.06187 ± 0.00254**
	REP	0.83716	0.65421	3.0	5.0	2.6	5.8	0.06192 ± 0.00221
	MEP	0.83716	0.65421	3.0	5.0	2.6	6.8	0.06203 ± 0.00183
	DBP	0.81614	0.68789	21.0	41.0	6.0	12.2	0.06261 ± 0.00220
QSARBiodeg	Base	0.81706	0.79614	101.2	201.4	15.4	30.4	0.07147 ± 0.00367
	CCP	0.84834	0.82883	38.2	75.4	10.2	18.8	0.06962 ± 0.00236
	Proposed SR-CCP	**0.85024**	**0.82896**	**5.8**	**10.6**	**4.4**	**4.6**	**0.06804 ± 0.00225**
	REP	0.83128	0.80521	14.2	27.4	8.8	17.4	0.06879 ± 0.00223
	MEP	0.84929	0.82760	17.6	34.2	9.2	19	0.06875 ± 0.00262
	DBP	0.83128	0.81160	25.6	50.2	6.0	16.8	0.06886 ± 0.00329
SpamBase	Base	0.90633	0.90212	239.4	477.8	29.0	46.6	0.14532 ± 0.00516
	CCP	**0.92154**	**0.91758**	71.4	141.8	16.2	30.2	0.12843 ± 0.00437
	Proposed SR-CCP	0.91784	0.91325	**6.4**	**11.8**	**4.6**	**5.2**	**0.11407 ± 0.00442**
	REP	0.90894	0.90382	20.2	39.4	10.4	23.2	0.11735 ± 0.00445
	MEP	0.91480	0.90981	24.8	48.6	11.4	29.2	0.11802 ± 0.00417
	DBP	0.90654	0.90064	23.4	45.8	6.0	14.2	0.11859 ± 0.00569

accuracy (86.55%) and exhibited greater variability across data sets, as shown in Figure 2.

Furthermore, for the BreastCancer, HeartDisease, and QSARBiodeg data sets, our proposed method obtained the best results. And for BanknoteAuth, Ionosphere, KC2, and SpamBase, our proposed SR-CCP reached the second-best results. Also, the best results for the all method were obtained for the BanknoteAuth data set, which has the minimum number of features and the minimum quantity of nodes, despite having a considerable number of samples. In addition, the greatest range of results is obtained in the Baseline database.

TABLE III. Summary of Mean and Median Accuracy and F1 Score by Pruning Method

Method	Accuracy		F1 score	
	Mean	Median	Mean	Median
Base	0.86549	0.89742	0.84464	0.88878
CCP	0.89218	0.91167	0.87456	0.90360
SR-CCP	**0.89467**	**0.91653**	**0.87722**	0.90618
REP	0.88011	0.90894	0.84498	0.90382
MEP	0.88806	0.91480	0.85347	**0.90981**
DBP	0.86960	0.89738	0.84418	0.88544

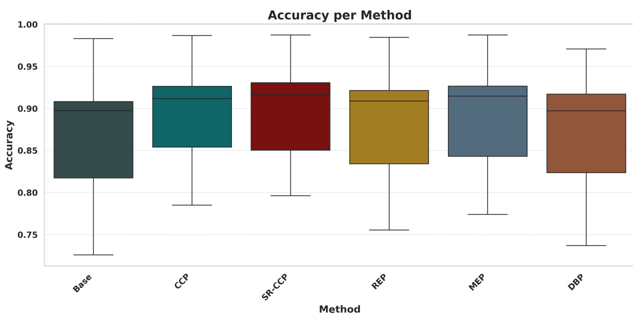

Figure 2. Box plot of accuracy across all data sets for each pruning method.

2) F1 score Analysis

Similarly, based on the results shown in Table II, the F1 score was calculated in the seven data sets for the six pruning strategies considered in this study, as shown in Figure 3 using box plot visualizations.

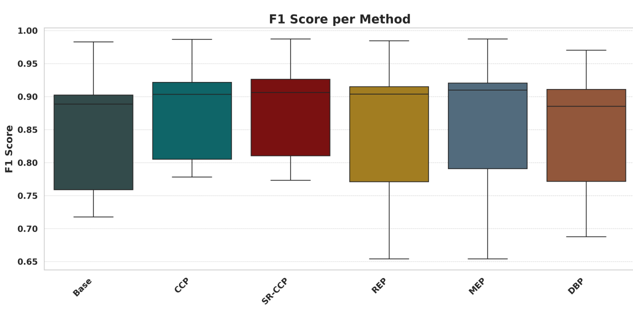

Figure 3. Boxplot of F1 score across all data sets for each pruning method.

Our proposed SR-CCP method achieved the highest overall F1 score, with a mean of 87.72% and a median of 90.62%, closely followed by CCP (87.46% mean, 90.36% median) and MEP (85.35% mean, 90.98% median). The unpruned Base model and the DBP methods produced a lower average F1 score (84.46% and 84.42%) and exhibited greater variability between data sets.

Particularly noteworthy are the results for:

- *Ionosphere*, where SR-CCP (90.62%) is competitive with CCP (90.36%) and MEP (91.34%).
- *KC2*, SR-CCP (77.33%) outperforms MEP/REP (both 65.42%) and DBP (68.79%).
- In *QSARBiodeg* SR-CCP reached the hightest F1 score with 82.90% and for *SpamBase*, SR-CCP shows the second-hightest value maintaining robust classification capability.

In addition, on the BanknoteAuth data set, the best results across all methods were again observed. And for the SpamBase data set, our proposed method shows good results despite this data set having the maximum number of features.

Figure 3 confirms that SR-CCP exhibits low dispersion and high consistency, reinforcing its robustness even in imbalanced settings.

3) Structural Complexity

The structural complexity of the decision trees was evaluated through four characteristics: the internal nodes, the terminal leaves, the maximum depth, and the number of features used.

Figure 4 presents a comparative analysis to the number of internal nodes generated by pruning strategies, including our proposed SR-CCP method. As can be seen, the SR-CCP consistently produces the most compact trees, achieving the lowest number of nodes in every data set. And the *SpamBase* method produces the highest number of nodes for all data sets. Comparing these two methods, the number of nodes is reduced from 477.8 (Base) to 11.8 (SR-CCP). Then REP and MEP follow closely related trends across all data sets, consistently generating models of intermediate size. DBP exhibits similar behavior in terms of node count to CCP, although it generally results in more compact models. CCP reduces node counts relative to the base model but does so less aggressively than other pruning techniques. Finally, we can show that the dispersion range of BanknoteAuth is less than SpamBase, due to the first data set only has 4 features, while the second one with 57 features.

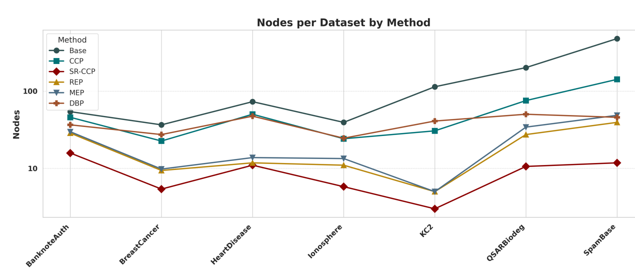

Figure 4. Number of nodes per data set and pruning method.

In addition, as is known in a binary tree, the number of leaves L is functionally related to the number of internal nodes N, and can be calculated using the equation 8.

$$Nodes = 2 * Leaves - 1 \tag{8}$$

Table IV shows the average number of nodes and leaves calculated per method in all data sets. SR-CCP produces the sparsest trees in both dimensions, while the base configuration results in the most complex structures.

TABLE IV. Mean number of nodes and leaves per method across all data sets.

Method	Nodes	Leaves
Base	142.3	71.7
CCP	55.8	28.4
DBP	39.0	20.0
MEP	22.1	11.5
REP	18.9	10.0
Proposed SR-CCP	9.1	5.0

Another important feature considered was the depth of the trees, illustrated in Figure 5. SR-CCP consistently produces the shallowest models, with maximum depths below 5.2 in all cases. REP and MEP follow almost identical trends, producing moderately shallow trees. DBP exhibits an invariant depth of 6 across all data sets, reflecting its design as a fixed depth constraint. CCP reduces depth relative to the base method, but its effectiveness varies by data set.

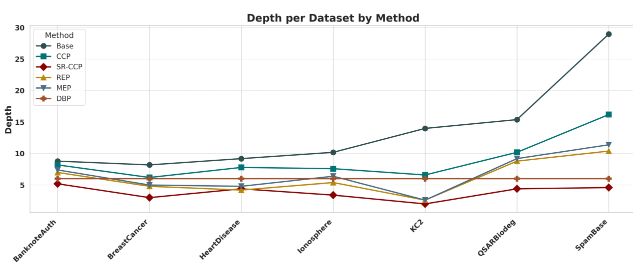

Figure 5. Maximum depth of the trees per data set and pruning method.

Figure 6 shows the number of features used in the structures of the final models. SR-CCP achieves the strongest feature regularization, consistently selecting a minimal subset of attributes. In *KC2*, for example, only a single feature is retained, compared to 19.2 in the unpruned baseline. REP and MEP follow similar patterns in feature selection. CCP, while less aggressive than SR-CCP, displays a feature usage trend that is closely aligned with MEP.

4) *Inference Time Analysis*

Figure 7 shows the average response time to the query (in milliseconds) to the average of 10,000 executions for each pruning method using the seven data sets and the standard deviation bars in the five stratified folds.

In general, all methods exhibit low and comparable inference times, with marginal differences in the data sets

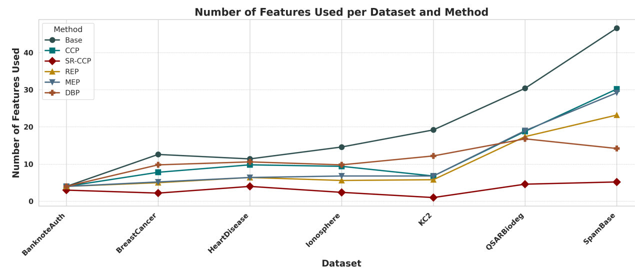

Figure 6. Number of features used per data set and pruning method.

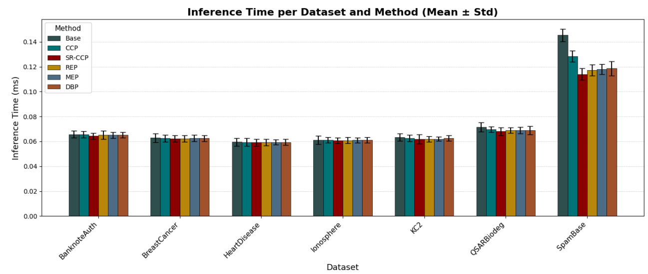

Figure 7. Mean inference time (ms) per method and data set, with standard deviation.

used in this study. The overall execution times remain below 0.07 ms except for the SpamBase data set, which exhibits higher latency due to its higher dimensionality and size. In addition to greater variability, it shows a standard deviation (SD) of 0.004 ms compared to 0.002 ms for the other data sets, which are shown in Table II.

However, notable distinctions emerge in the QSARBiodeg and SpamBase data sets. In these cases, SR-CCP demonstrated a clear advantage, achieving the lowest inference times among all methods. Specifically, in SpamBase, SR-CCP reduces inference latency by more than 21% compared to the Base model (0.114ms vs. 0.145ms), with a similarly favorable reduction in QSARBiodeg (0.068 ms vs. 0.071 ms in Base). These improvements were achieved without increasing variability and while maintaining a standard deviation comparable to other methods.

Overall, this behavior suggests that, while pruning methods generally do not introduce significant inference overhead, the proposed SR-CCP can offer measurable performance benefits in more complex or high-dimensional tasks.

5) *Multivariate Evaluation*

A radar plot [26] is a type of graphical representation that is used to explore and display patterns in the data. It is considered a tool for building confidence in a single information trustworthiness metric [27].

In our case, to provide a multivariate evaluation of pruning strategies, radar plots were constructed for each data set. As shown in Figure 8. Each plot is visualized as a whole normalized performance dimension, including accuracy, F1 score, tree depth, number of nodes, number of leaves, number of features used, and inference time. All values were normalized to the interval $[0, 1]$ such that higher values uniformly indicate better outcomes. Also, for the BanknoteAuth plot, the proposed SR-CCP displays Accuracy at 98.74%, F1 score at 98.73%, Leaves at 85.66%, Nodes at 85.66%, Depth at 76.40%, Features at 50%, and Time Query at 72%. Shows the best performance compared to the other method. Therefore, this direct visual comparison uses all metrics, where larger closed areas indicate methods that perform better in both the predictive and structural dimensions.

Across all data sets, the SR-CCP method consistently produces the broadest and most balanced radar profiles, achieving competitive predictive accuracy while minimizing structural complexity, as demonstrated by its performance on data sets such as *HeartDisease*, *KC2*, and *SpamBase*. These results confirm the effectiveness of the proposed structural regularization pruning scheme, SR-CCP, in generating accurate and parsimonious decision trees.

In contrast, the Base model forms irregular and contracted radar shapes, particularly in structural dimensions such as Depth, Nodes, and Leaves. These patterns reflect its high structural complexity due to the absence of pruning, resulting in deeper and larger trees that are harder to interpret and more prone to overfitting.

The pruning strategies REP and MEP exhibit notably similar behavior across all data sets. Both methods generate models with moderate improvements over the Base method, but neither outperforms CCP or SR-CCP in structural or predictive terms. Their radar shapes frequently overlap and demonstrate only partial gains, reinforcing their limited advantage in complex or noisy data sets.

Depth pruning shows a consistent but limited performance profile. The proposed SR-CCP achieves competitive results in terms of tree compactness, with robust values for depth, nodes, and leaves. Many experiments were conducted using seven data sets proving that it outperforms all in structural dimensions and predictive metrics (accuracy and F1 score). Thus, the resulting radar plots show broader profiles along the structural axes and only slight compression along the performance axes, indicating that the method successfully balances simplification with high predictive power.

Furthermore, the Inference Time metric exhibits minimal variation between methods and data sets. Although SR-CCP shows slightly better latency due to reduced tree complexity, all methods achieve similar values along this axis, suggesting that inference efficiency is not a primary differentiator among the strategies.

5. Conclusion

This study proposes a new method, called SR-CCP Pruning, as an extension of the CCP method, that incorporates explicit penalties on depth and the proportion of

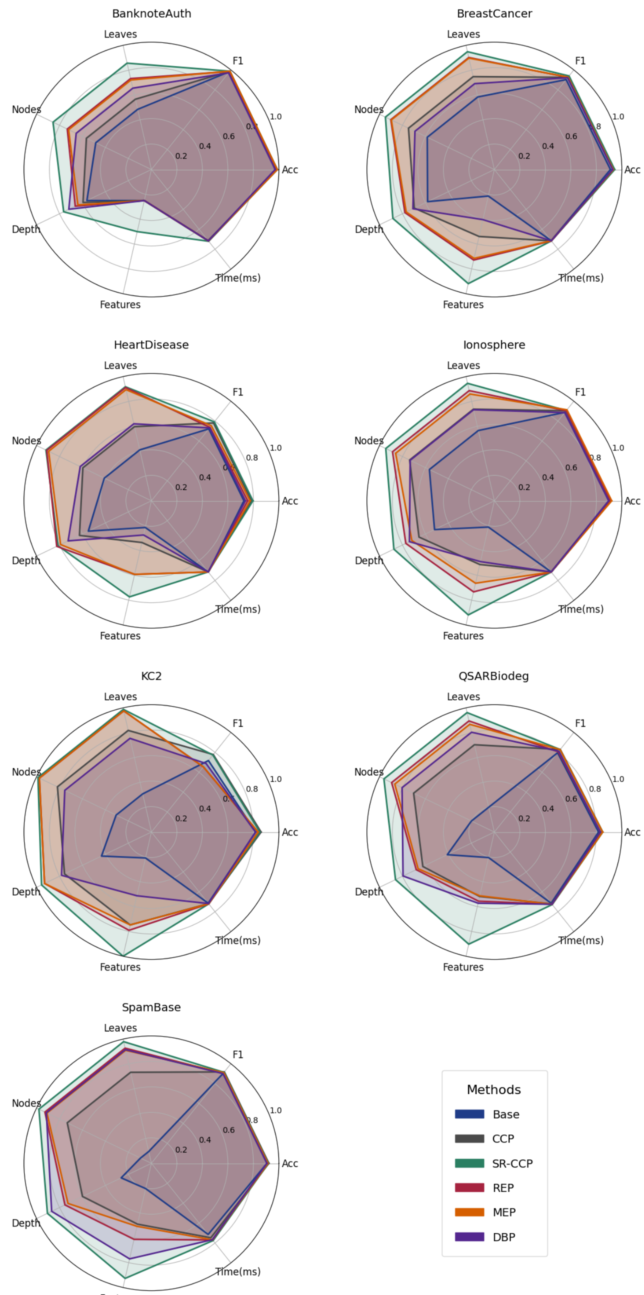

Figure 8. Normalized radar plots across seven evaluation metrics for each data set.

features used during pruning, whose results are shown in a comparative analysis of six pruning strategies applied to decision tree classifiers, evaluated across seven data sets from heterogeneous application domains. The assessment incorporated a broad spectrum of performance metrics, encompassing both predictive accuracy and structural complexity, including cross-validated accuracy, F1 score, tree depth, number of internal nodes, terminal leaves, feature utilization, and inference latency. Among the methods examined, the proposed SR-CCP algorithm emerged as a robust and effective alternative.

Empirical results demonstrated that SR-CCP consistently maintains competitive predictive performance across all pruning methods, preserving comparable accuracy and F1 score values in most data sets. Importantly, SR-CCP achieved substantially greater structural parsimony, producing significantly more compact trees characterized by reduced depth, fewer internal nodes, and a lower number of terminal leaves compared to all other variants of the pruning model. MEP and REP exhibited similar behavior and occasionally approached SR-CCP in structural and feature usage metrics, but did not exceed it.

Radar plot visualizations further substantiated these findings. Base models produced smaller and more irregular polygonal shapes, indicating greater structural complexity and less balanced performance across metrics, whereas SR-CCP consistently generated larger, more regular, and balanced polygons, reflecting effective structural constraints and enhanced model parsimony. Regarding inference time, all pruning methods exhibited broadly similar computational costs, with SR-CCP showing marginally faster inference on the QSARBiodeg and Spambase data sets.

A principal contribution of SR-CCP is its ability to enforce structural simplification, which is particularly advantageous in scenarios demanding interpretable, efficient, and resource-constrained models, such as embedded systems, biomedical monitoring devices, and human-in-the-loop decision frameworks. Moreover, despite the strong structural regularization imposed by SR-CCP, the method maintained and often enhanced predictive performance, achieving superior or comparable results in accuracy and F1-score across most data sets. This demonstrates that the pursuit of extreme simplicity with SR-CCP does not compromise predictive capability, making it a robust choice even in performance-critical applications such as medical diagnosis or financial risk assessment.

Future research directions include integrating SR-CCP within ensemble learning frameworks, such as Random Forests and Gradient Boosted Trees, to investigate whether structural regularization can enhance generalization and interpretability at the ensemble level. Moreover, the development of adaptive penalization mechanisms that adjust dynamically based on data set characteristics-such as dimensionality or noise-could further improve method flexibility. Expanding evaluation to include time-series classification, multi-label problems, and streaming data environments also represents a promising avenue for assessing the broader applicability of SR-CCP across diverse machine learning tasks.

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